

Multimodal Fashion & Context Retrieval — Write-up

Glance ML internship assignment. Code: see [GitHub link](#) in §4.

1. Approaches considered & trade-offs

The task looks like "just use CLIP", but vanilla CLIP has two documented failure modes that this assignment's queries deliberately probe:

- Compositionality blindness.** CLIP embeds an image as a *bag of concepts*: "red shirt with blue pants" and "blue shirt with red pants" land at nearly the same point because attribute→object *binding* is not preserved in a single pooled vector.
- Fine-grained fashion weakness.** Generic CLIP training data underspecifies fashion: blazer vs cardigan, fabric, fit and formality nuances blur together.

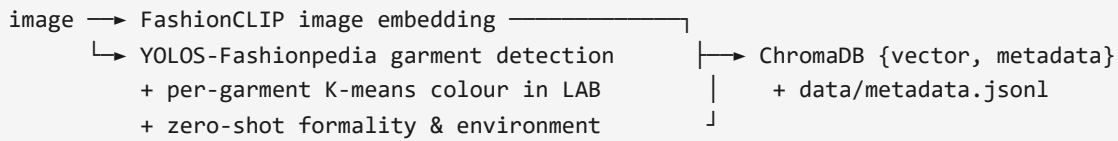
Every candidate architecture trades off along **recall vs precision vs zero-shot generality**:

#	Approach	Recall	Binding precision	Zero-shot	Verdict
a	Vanilla CLIP, dense-only	high	poor — bag-of-concepts	high	Baseline; fails queries 1 & 5
b	FashionCLIP, dense-only	high	poor — same pooling, better features	high	Fixes fine-grained, not binding
c	CLIP + hard metadata filter	low — a single wrong detection/label removes a relevant image irrecoverably	high when metadata is right	medium	Brittle; detector errors are fatal
d	Hybrid retrieve-then-rerank (chosen)	high (dense stage untouched)	high (structured binding score)	high (falls back to cosine when attributes are silent)	Best of both; soft scores degrade gracefully
e	Fully structured / tag search (no embeddings)	very low — only what the tag vocabulary covers	high	none — "casual weekend outfit" has no tag	Cannot do style inference

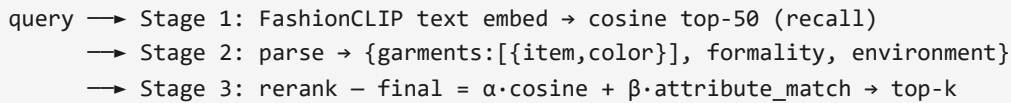
When each wins: (a)/(b) win when queries are single-concept ("a dress"); (c) wins when metadata is human-curated and exact filtering is a product requirement; (e) wins in closed catalogues with clean structured data. For open-ended natural-language queries over noisy images — this assignment — (d) dominates because errors in either signal are *soft*: a missed detection costs some rerank score instead of eliminating the image, and a fuzzy query still gets full dense recall.

2. Chosen architecture

INDEXER (Part A)



RETRIEVER (Part B)



Why FashionCLIP (patrickjohncyh/fashion-clip): CLIP fine-tuned on 800K fashion image-text pairs. It directly attacks failure mode #2 (fine-grained garments) at zero engineering cost, and keeps the zero-shot interface we need for formality/environment prompts. It does *not* fix binding — that is the reranker's job.

Why each stage exists:

- **Stage 1 — dense recall.** Cheap wide net; anything semantically close survives to the rerank. Zero-shot generality lives here: "casual weekend outfit" needs no tag or label to recall the right neighbourhood.
- **Stage 2 — structured parsing.** *This is where compositionality is solved.* Binding is lifted out of the smeared embedding into explicit structure: [{item: tie, color: red}, {item: shirt, color: white}]. Three tiers, each a strict fallback for the last: a strict-JSON **LLM** call; an offline **dependency parse** (spaCy) that reads the binding off the syntax tree; and a zero-dependency **adjacency rule** ("a colour binds to the nearest *following* garment noun" — the English pre-nominal adjective rule). The system passes all five eval queries and the swap test with the LLM disabled.

Why the syntax tier exists (and why not attention). The adjacency rule is right on pre-nominal colour and silently wrong everywhere else: measured on a 19-construction suite it scored **5/19**, dropping the colour entirely on "a shirt that's red" / "a shirt in red" / "the shirt is white" (the clause degrades to category-only, so *any*-coloured shirt scores 1.0 — the binding machinery quietly switches off) and mis-binding coordination ("a red and white shirt" → white; "a red shirt and tie" → colourless tie). The obvious fix is to read the binding out of CLIP's own attention — but that is *circular*, and measurably useless: CLIP's text encoder is causal (a colour token cannot attend forward to its noun at all — weight 0.000), and its backward attention is **positional, not lexical**: swapping the colours in "a shirt in red and pants in blue" moves the weights by <0.003. It would reproduce the adjacency heuristic, failures included. Syntax is the right instrument for syntax: a dependency parse maps `amod/acomp/reicl/ prep+pobj/conj` onto exactly the bindings we need, offline and deterministically — **19/19** on the construction suite. Note this bug was invisible to the benchmark (all five eval queries are pre-nominal), which is why that suite is now a regression test in its own right.

...but the syntax tier alone is not enough either, and this is why the parser routes. Dependency parsers assume **grammatical** input. Real search queries often are not: "red shirt blue pants", "navy blazer grey trousers" have no verb and no determiners, and the small spaCy model reads them as a single compound noun-chain (`shirt -- compound--> pants`), collapsing the garments and re-attaching colours to the wrong nouns. On such inputs the syntax tier scores **8/12** — **worse than the adjacency rule's 9/12**, and its errors are *active mis-bindings* rather than silent drops. So the two offline parsers fail on **disjoint** inputs, and `parse()` arbitrates: if the tree recovers fewer garments (or fewer bound colours) than a plain lexical scan, the tree has collapsed and adjacency wins. Routing scores **18/19** overall, beating either parser alone (15/19 syntax, 14/19 adjacency).

Would a transformer parser be better? Measured, rather than assumed — and the answer is yes, at a price. Swapping spaCy's convolutional encoder for its RoBERTa one (en_core_web_trf) scores **19/19 with no router at all**, including 12/12 on the telegraphic fragments the small model mangles. It costs **440 MB and ~40 ms/query against 12 MB and ~4 ms**. The default stays on the small model, because the router buys back all but one case for 3% of the disk; `models.spacy` in `config.yaml` trades up in one line. This is the honest shape of the trade — a cheap model plus a cheap fallback rule beats a model 37× its size on 18 of 19 cases.

Formality/environment additionally get a **query-side zero-shot classification** (the query embedding scored against the same prompt banks as images), fused with the keyword parse via a three-band trust ladder: unconfident → keyword only (axis drops if keywords found nothing, so unspecified never locks on); moderately confident → explicit keywords win conflicts; highly confident → the embedding wins, which corrects keyword false-positives ("smart *casual*" no longer reads as formal via the keyword "smart") and recovers axes keywords miss entirely ("hiking by a lake" → park at 0.97 confidence). - **Stage 3 — attribute rerank**. *This is where precision is won*. Per parsed clause, the **binding score** rewards a candidate only if some single detected garment matches **both** category and colour: an image with a blue tie and a red shirt earns almost nothing for "red tie". Colour matching is continuous in LAB space (perceptually uniform, so $\Delta E \approx$ human colour difference): the query word is grounded to its own LAB point via a ~45-word colour lexicon, so *maroon* is a target near-but-not-equal-to red, *navy* earns ~0.6 of a *blue* query's credit, and *white* bottoms out near zero. An out-of-palette word (*burgundy*, *teal*) therefore stays a constraint instead of silently vanishing. Formality and environment use the stored zero-shot **probabilities**, not argmax labels, so a near-tie classification is not punished like a confident mismatch. Axes the query never mentions are excluded from the weight normalisation — unspecified never penalises. Final score: $\alpha \cdot \text{cosine} + \beta \cdot \text{attr}$ ($\alpha=0.45$, $\beta=0.55$, weights in `config.yaml`).

Two design details worth calling out:

- **Per-garment colour, not per-image colour**. Dominant colour is K-means (k=3) over the *detected garment's crop* in LAB. Whole-image colour would reintroduce exactly the bag-of-concepts failure we are fixing.
- **Greedy one-to-one clause matching**. One white shirt cannot satisfy both "white shirt" and "white pants" — each detection pays for at most one clause.

How the five assignment queries flow through it:

Query	Dense stage contributes	Structured stage contributes
1. "bright yellow raincoat"	raincoat-like images	parser: (coat, yellow) + outerwear; binding score demands yellow <i>on the coat</i>
2. "business attire inside a modern office"	suits/formal looks	formality=formal (prob.), environment=office (prob.)
3. "blue shirt sitting on a park bench"	people outdoors in shirts	binding (shirt, blue) + environment=park
4. "casual weekend outfit for a city walk"	carries the query — pure style inference, no literal labels	formality=casual, environment=street nudge the top
5. "red tie and white shirt, formal"	ties/shirts/formal wear	the differentiator : two clauses, each must bind on its own garment

3. Results

Corpus: the full 3,200-image Fashionpedia val_test2020 pool, fully indexed (the system was developed on a 693-image subsample, then scaled up and re-tuned against full-corpus error analysis). Relevance: hand-labelled by pooled evaluation — for each query the union of top-10 results from all three ablation systems was judged visually; criteria per query in `eval/relevance_criteria.md`. Grids: `eval/results/*.png`.

Hybrid system on the five assignment queries (k=5)

Query	P@5	R@5	Notes
1 Attribute (yellow raincoat)	0.80	0.67	binding demands yellow <i>on the coat</i>
2 Contextual (office)	0.80	0.20	formality+environment soft scores
3 Complex (blue shirt, park bench)	0.40	0.33	see failure analysis below
4 Style (casual weekend)	1.00	0.14	dense zero-shot carries it
5 Compositional (red tie, white shirt)	0.80	0.40	corpus caveat below
Mean	0.76	0.35	

Ablation — P@5, all three systems, all five queries (full corpus)

Query	Vanilla CLIP dense	FashionCLIP dense	Hybrid (ours)
attribute	0.60	0.60	0.80
contextual	0.40	1.00	0.80
complex	0.20	0.40	0.40
style	0.80	1.00	1.00
compositional	0.80	0.40	0.80
Mean	0.56	0.68	0.76

Vanilla→FashionCLIP isolates the fine-grained-fashion gain (+0.12); FashionCLIP→hybrid isolates the structured-attribute gain (+0.08 mean, +0.20 on the attribute query where colour binding decides, and +0.40 on the compositional query, where FashionCLIP's dense ranking actually *trails vanilla* — better fashion features do not fix binding; structure does).

Corpus caveat on query 5, and how we measured binding anyway. An exhaustive FashionCLIP probe over the full 3,200-image pool found **no genuine red necktie** — vanilla CLIP's best "red tie" matches pool-wide are red *dresses* (the attribute leaks to the wrong garment — the bag-of-concepts failure in one picture, `eval/pool/` shows it). Relevance for query 5 was therefore relaxed to "necktie + white shirt + formal", which removes the colour-binding signal and lets a dense suit-retriever tie the hybrid. So we added a **swap-pair binding probe** on a colour pair the corpus *does* contain, with a strong natural asymmetry: ~23 white-top+black-pants images vs exactly **one** verified black-top+white-pants image. Querying the rare direction floods a bag-of-concepts retriever with swapped matches:

(At full corpus the rare direction has 4 verified true matches among ~26 white-top+black-pants images.)

System	"black shirt + white pants" (rare): P@5 and true-match ranks	"white shirt + black pants" (control): P@5
Vanilla CLIP dense	0.00 — best true match at #10	0.40
FashionCLIP dense	0.40 — ranks #1, #2, #8	0.60
Hybrid (ours)	0.60 — ranks #1, #3, #4	1.00

This is the compositionality result in its purest measurable form on this corpus: same words, opposite binding — only the hybrid reliably separates them. The swap unit test (`tests/test_compositionality.py`) proves the same mechanism analytically: identical-cosine synthetic images with swapped shirt/pants colours rank in opposite orders under the two swapped queries.

Failure analysis — query 3 (blue shirt + park bench). The hardest query: a garment clause AND a scene clause, multiplying both subsystems' error rates. Full-corpus error analysis drove three measured fixes (each A/B'd, two other candidate fixes were rejected by the data — see below): after them, the four best-detected relevant images rank #1, #3, #6, #8. What remains is instructive: the top false positives are a real navy shirt failing only the un-modelled *sitting* posture, and denim tees on the street; the missed relevant images are detector failures (a blue jersey whose K-means colour reads brown, a plaid shirt averaging to gray). Pose modelling, segmentation-based colour, and a real scene classifier (future work) target exactly these residuals.

What full-corpus error analysis changed — and what it refused to change. Scaling from 693 to 3,200 images was used as a tuning pass, one lever at a time with the eval re-run after each: 1. **Achromatic gate (added):** black shirts (measured chroma 5-8) were earning 0.30 credit toward "blue" — as much as a genuinely bluish-gray tee. A garment with no chroma now earns only the floor toward a chromatic query. Black-shirt false positives left the complex top-5; no blue-hued garment (navy 24, bluish-gray 15) was touched. 2. **Absent-credit 0.3 → 0.2 (adjusted):** garmentless park photos were riding the environment score past true matches because "no garment detected" scored the same as real-but-imperfect colour evidence. 3. **Colour ramp tightening (rejected):** narrowing `color_soft_max` 100→80 killed the beige→yellow leak but dropped complex P@5 from 0.40 to 0.20 — the query's true matches are navy/denim tops living exactly in the punished delta-E band. Reverted. 4. **Wider dense recall (rejected):** doubling top-N to 100 *hurt* (complex 0.40→0.20). The misses were never recall failures — dense already ranked them #2/#5/#6 — they were reranker demotions of detector-blind images, and the wider pool only imported extra distractors.

Error trade-off we accepted. The garment detector runs at a deliberately low confidence threshold (0.25): a missed garment silently destroys binding evidence, while a spurious detection is absorbed by soft scoring (wrong-colour matches floor out near zero). The reranker encodes the matching evidence hierarchy explicitly: correct binding (1.0) >> perceptual near-miss (e.g. navy for "blue" $\approx$ 0.6) >> category never detected (0.30, weak evidence) > detected with contradicting colour (0.05, strong counter-evidence). The colour ramp is calibrated against palette-anchor spacing: its decay length (delta-E 100) is about twice the typical distance between neighbouring named colours, so neighbours score partial credit while opposites hit the floor — with a decay length below that spacing, soft matching would degenerate to exact-name-or-floor.

4. Code

GitHub: <https://github.com/Varshith-06/GlanceAssignment> — README.md has setup, run commands and the architecture diagram. Runs CPU-only, offline (LLM parser optional). `pip install -r requirements.txt && python indexer/build_index.py` rebuilds everything from scratch; all stages are idempotent.

5. Future work

Locations & weather. Add a dedicated scene classifier (Places365) rather than CLIP prompts → richer environment embedding + city metadata as filterable fields; add weather tags (rainy/sunny/snowy) either from image classification or paired EXIF/caption data; extend the query parser to emit `{location, weather}` slots that feed the same reranker — the scoring framework already generalises to new axes by adding a weight. (We prototyped the weather axis: it wires in exactly as described, but on this corpus visual weather cues appear in <1% of photos, and weather *inferred* from garment words — "yellow raincoat" → rainy at 0.92 confidence — penalised correct garment matches shot in dry conditions. The axis pays off only with data that actually encodes weather, e.g. EXIF/caption pairing, so it stays future work.)

Precision. (1) Fine-tune FashionCLIP with **hard negatives**: auto-generate colour-swapped caption pairs ("red tie, white shirt" ↔ "white tie, red shirt") and train contrastively — this attacks compositionality *at the embedding level* instead of patching it post-hoc. (2) Add a cross-encoder reranker (e.g. BLIP-2 ITM) over the final top-k — joint attention over image+text sees binding natively; affordable because k is small. (3) Replace box crops with garment *segmentation* masks before LAB colour naming — removes background bleed, the main colour-extraction error source.

Scaling to 1M images. Swap Chroma's flat search for FAISS IVF/HNSW — sub-linear ANN keeps the dense stage $\sim O(\log n)$; pre-filter candidates by structured metadata (e.g. `environment=office`) before ranking to shrink the candidate set; batch/GPU embedding for indexing throughput. The key architectural property is already in place: **only the top ~50 candidates are ever reranked, so rerank cost is constant regardless of corpus size** — the expensive precision machinery never touches the full corpus.